synopsys®

DesignWare® Cores
SuperSpeed USB 3.0 Controller

Documentation Overview

DWC_usb3 – Product Codes

Version 3.30b
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Documentation Overview

This document provides you with a complete listing of the documentation that ships with your Synopsys controller product. It also provides directions for you to find Synopsys Common Licensing information and other commonly used information.

1.1 Product Codes

Table 1-1 lists the products and product codes for the DWC_usb3 controller.

Table 1-1 Product Codes for DWC SuperSpeed USB 3.0 Controller

Component Name or Add-On	Product Code
Base Products	
DWC USB 3.0 Device-AHB[1]	6793-0
DWC USB 3.0 Host-AHB[2]	6897-0
DWC USB 3.0 Dual Role Device-AHB[3]	7567-0
DWC USB 3.0 Hub	6921-0
DWC AP USB 3.0 Device	C234-0
DWC AP USB 3.0 Host	C233-0
DWC AP USB 3.0 Dual Role Device	C232-0
Add-Ons	
DWC USB 3.0 Device-Isoch Add-On	7571-0
DWC USB 3.0 Host-Isoch Add-On	7593-0
DWC USB 3.0 DRD-Isoch Add-On	7568-0
DWC USB 3.0 Hibernation (LCA) Add-On	9228-0
DWC USB 3.0 HSIC Add-On	7387-0

1. DWC USB 3.0 Device-AHB includes DWC USB 3.0 Device-AXI Add-On and DWC USB 3.0 Device-Isoch Add-On.
2. DWC USB 3.0 Host-AHB includes DWC USB 3.0 Host-AXI Add-On and DWC USB 3.0 Host-Isoch Add-On.
3. DWC USB 3.0 Dual Role Device-AHB includes DWC USB 3.0 DRD-AXI Add-On and DWC USB 3.0 DRD-Isoch Add-On.

1.2 SuperSpeed USB 3.0 Controller Documentation

The SuperSpeed USB 3.0 Controller documentation set comprises the following documents:

- Full documentation set that includes information on all modes of operation (host, device, DRD, and hub):
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Documentation Overview* (this document)
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Release Notes*
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Installation Guide*
  - *DesignWare Cores SuperSpeed USB 3.0 Controller User Guide*
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Databook*
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Databook (Changes)*
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Programming Guide*
  - *DesignWare Cores SuperSpeed USB 3.0 Controller Programming Guide (Changes)*
- Host-specific documentation that includes information on host mode of operation only:
  - *DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller User Guide*
  - *DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller Databook*
  - *DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller Databook (Changes)*
  - *DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller Programming Guide*
  - *DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller Programming Guide (Changes)*

Table 1-2 USB 3.0 Controller Document List

General Documents	Description
Full Documentation Set	
DesignWare Cores SuperSpeed USB 3.0 Controller Documentation Overview	This document describes, with links, documentation in this doc set.
DesignWare Cores SuperSpeed USB 3.0 Controller Release Notes	Lists new features in the release, problems fixed in the release, and known problems and limitations.
DesignWare Cores SuperSpeed USB 3.0 Controller Installation Guide	Provides product installation information, including product licensing information, hardware and operating system requirements, required environment variables, and download and setup instructions.
DesignWare Cores SuperSpeed USB 3.0 Controller User Guide	Includes information on how to configure the controller, verify, and integrate it into your chip design.
DesignWare Cores SuperSpeed USB 3.0 Controller Databook	Provides a complete description of the USB 3.0 Controller, including compliance information and features, configuration parameters, and signal descriptions, architecture, clocks, and resets.
DesignWare Cores SuperSpeed USB 3.0 Controller Databook (Changes)	A version of the Databook with change bars indicating information that has changed from the previous release. Wording and formatting changes that do not affect functionality are not marked.
DesignWare Cores SuperSpeed USB 3.0 Controller Programming Guide	Provides the programming information of the USB 3.0 Controller in device and host mode including the register descriptions. It also describes the programming flow for various features.
DesignWare Cores SuperSpeed USB 3.0 Controller *Programming Guide* (Changes)	A version of the Programming Guide with change bars indicating information that has changed from the previous release. Wording and formatting changes that do not affect functionality are not marked.
Host-Specific Documentation Set	
DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller User Guide	Includes information on how to configure the host-only controller, verify, and integrate it into your chip design.
DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller Databook	Provides a complete description of the USB 3.0 xHCI Host Controller, including compliance information and features, configuration parameters, signal and register descriptions, architecture, clocks, and resets.
DesignWare Cores SuperSpeed USB 3.0 xHCI Host Databook (Changes)	A version of the USB 3.0 xHCI Host Controller Databook with change bars indicating information that has changed from the previous release. Wording and formatting changes that do not affect functionality are not marked.
DesignWare Cores SuperSpeed USB 3.0 xHCI Host Controller *Programming Guide*	Provides the programming information of the USB 3.0 xHCI Host Controller including the register descriptions. It also describes the programming flow for various features.
DesignWare Cores SuperSpeed USB 3.0 xHCI Host *Programming Guide* (Changes)	A version of the USB 3.0 xHCI Host Controller Programming Guide with change bars indicating information that has changed from the previous release. Wording and formatting changes that do not affect functionality are not marked.

After product installation, all documents are installed in the following locations:

install_dir/iip/DWC_usb3/latest/doc

After you complete the Specify Configuration activity in coreConsultant, documents are available at the following locations:

- *workspace*/doc/ directory
- coreConsultant Help Menu

1.3 Installing and Configuring the Controller

Figure 1-1 shows the usage flow for installation and configuration tasks and subtasks of the DWC_usb3 controller.

Figure 1-1 Installation and Configuration Tasks

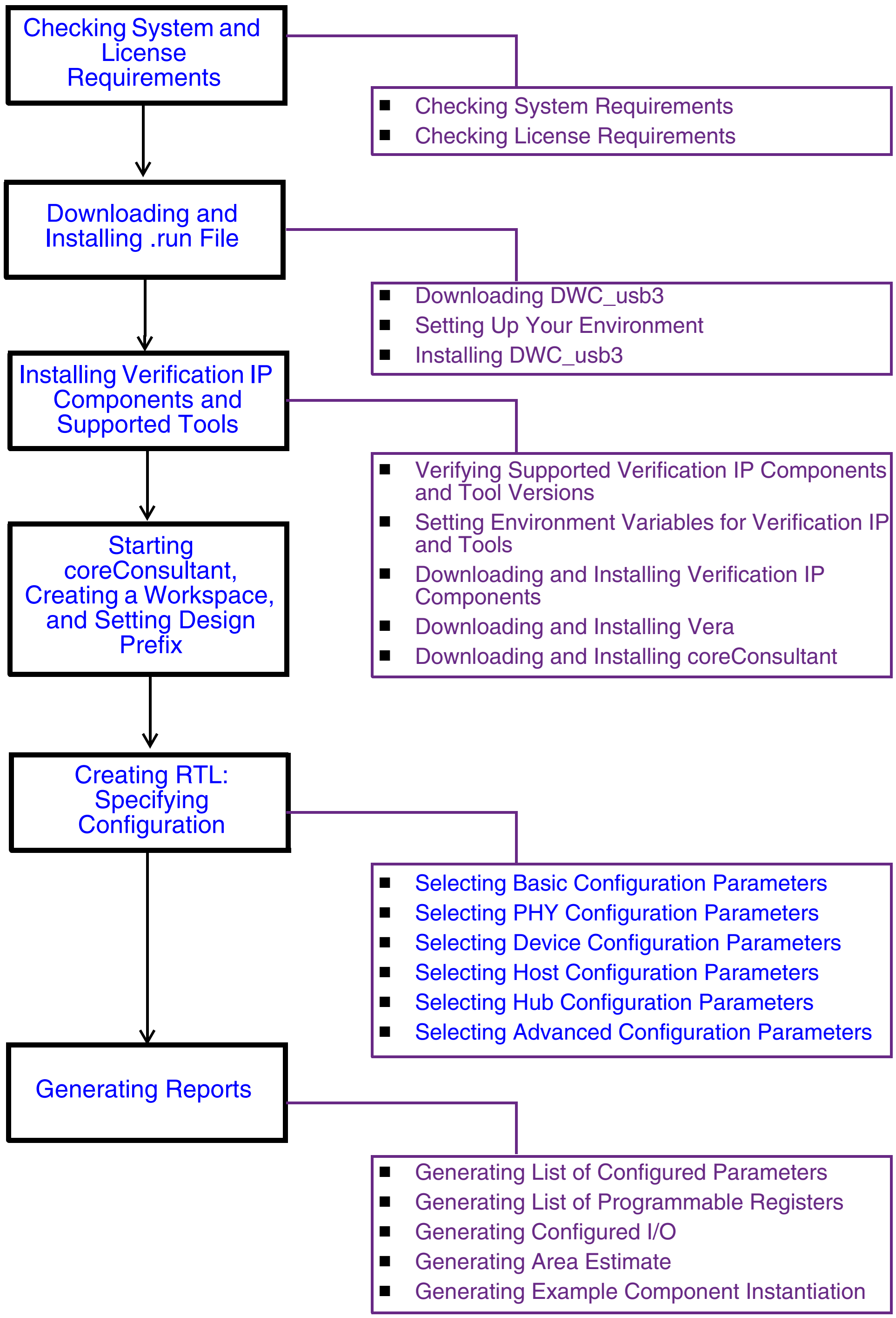

Table 1-3 provides the installation and configuration tasks and the respective documentation locations:

Table 1-3 Installation and Configuration Tasks

Task	Description	Document Location
Checking System and License Requirements	Check system and license requirements.	*Installation Guide* ■ "Checking System Requirements" ■ "Checking License Requirements"
Downloading and Installing .run File	Download, set up the environment variables, and install DWC_usb3.	*Installation Guide* ■ "Downloading DWC_usb3" ■ "Setting Up Your Environment" ■ "Installing DWC_usb3"
Installing Verification IP Components and Supported Tools	Download and install Verification IP components and supported tools.	*Installation Guide* ■ "Verifying Supported Verification IP Components and Tool Versions" ■ "Setting Environment Variables for Verification IP and Tools" ■ "Downloading and Installing Verification IP Components" ■ "Downloading and Installing coreConsultant"
Starting coreConsultant, Creating a Workspace, and Setting Design Prefix	Use the Synopsys coreConsultant tool to configure, simulate, synthesize, and export DWC_usb3 to your design flow. Create a workspace, which is a local Unix directory structure containing your configured copy of the DWC_usb3.	*Installation Guide* ■ "Getting Started with coreConsultant" Chapter 2 of the *User Guide*: ■ "Setting up your Environment" ■ "Creating a Workspace"
Creating RTL: Specifying Configuration	Generate the RTL by specifying the values for various parameters and by selecting the required features.	"Configuring the Controller" in Chapter 2 of the *User Guide* For information on the subtasks, see Table 1-4 on page 9.
Generating Reports	You can generate and view various reports (some optional).	"Creating Optional Reports and Views" in Chapter 2 of the *User Guide*

Table 1-4 Specifying Configuration Subtasks

Task	Description	Document Location
Selecting Basic Configuration Parameters	In coreConsultant, select the basic configuration parameters suitable for your application.	"Basic Config Parameters" section in Chapter 4, Parameter Descriptions of *Databook*
Selecting PHY Configuration Parameters	In coreConsultant, select the PHY configuration parameters suitable for your application.	"PHY Config Parameters" section in Chapter 4, Parameter Descriptions of *Databook*
Selecting Device Configuration Parameters	If you are designing a USB device or Dual Role Device (DRD), in coreConsultant, select the device configuration parameters (in coreConsultant) suitable for your application.	"Device Config Parameters" section in Chapter 4, Parameter Descriptions of *Databook*
Selecting Host Configuration Parameters	If you are designing a USB host or DRD, select the host configuration parameters (in coreConsultant) suitable for your application.	"Host Config Parameters" section in Chapter 4, Parameter Descriptions of *Databook*
Selecting Hub Configuration Parameters	If you are designing a USB hub, select the hub configuration parameters (in coreConsultant) suitable for your application.	"Hub Config Parameters" section in Chapter 4, Parameter Descriptions of *Databook*
Selecting Advanced Configuration Parameters	In coreConsultant, select the advanced configuration parameters suitable for your application.	"Advanced Config Parameters" section in Chapter 4, Parameter Descriptions of *Databook*

1.4 Verifying the Controller

Figure 1-2 shows the usage flow for simulation tasks and subtasks of the DWC_usb3 controller.

Figure 1-2 Controller Simulation Tasks

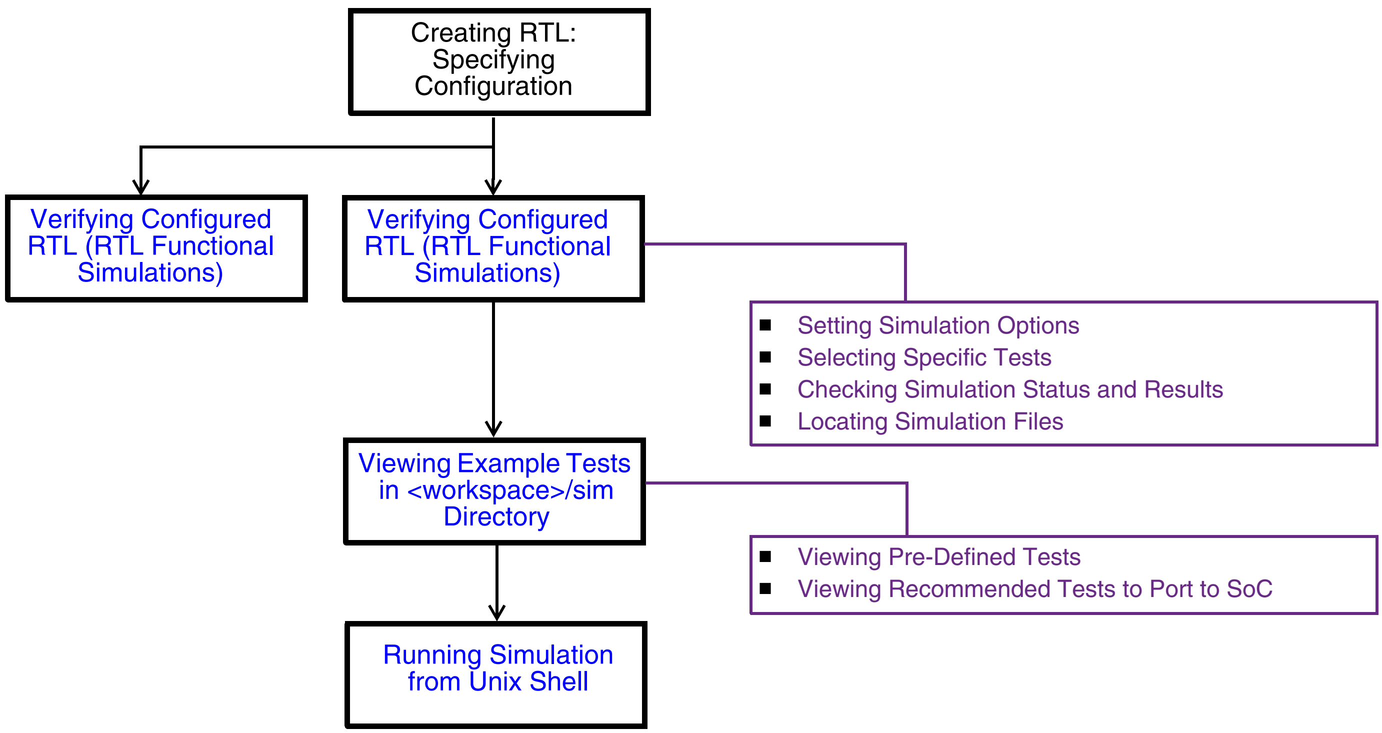

Table 1-5 provides controller simulation tasks and the respective documentation locations:

Table 1-5 Controller Simulation Tasks

Task	Description	Document Location
Verifying Configured RTL (RTL Functional Simulations)	Simulate the generated RTL in the Packaged Verification Environment (PVE) provided with the DWC_usb3.	"Running PVE Tests Using coreConsultant" in Chapter 3 of *User Guide*
Viewing Example Tests in <*workspace*>/sim Directory	View and understand the tests. The Packaged Verification Environment (PVE) used internally by Synopsys to verify the DWC_usb3 controller. Synopsys provides this PVE with the DWC_usb3 controller so that you can run a basic test suite on your configured controller in a closed environment.	"PVE Test Bench" in Chapter 3 of *User Guide*
Running Simulation from Unix Shell	Manually execute the run scripts from the UNIX command line.	"Running PVE Tests from Command Line" in Chapter 3 of *User Guide*
Running VCS Xprop Analyzer	Optional. Run the Xprop feature of the VCS simulator in the coreConsultant environment to check for the instrumentation capabilities of Xprop on the generated RTL.	"Running VCS Xprop Analyzer" in Chapter 3 of *User Guide*

1.5 Synthesizing the Controller

Figure 1-3 shows the usage flow for synthesis tasks and subtasks of the DWC_usb3 controller.

Figure 1-3 Controller Synthesis Tasks

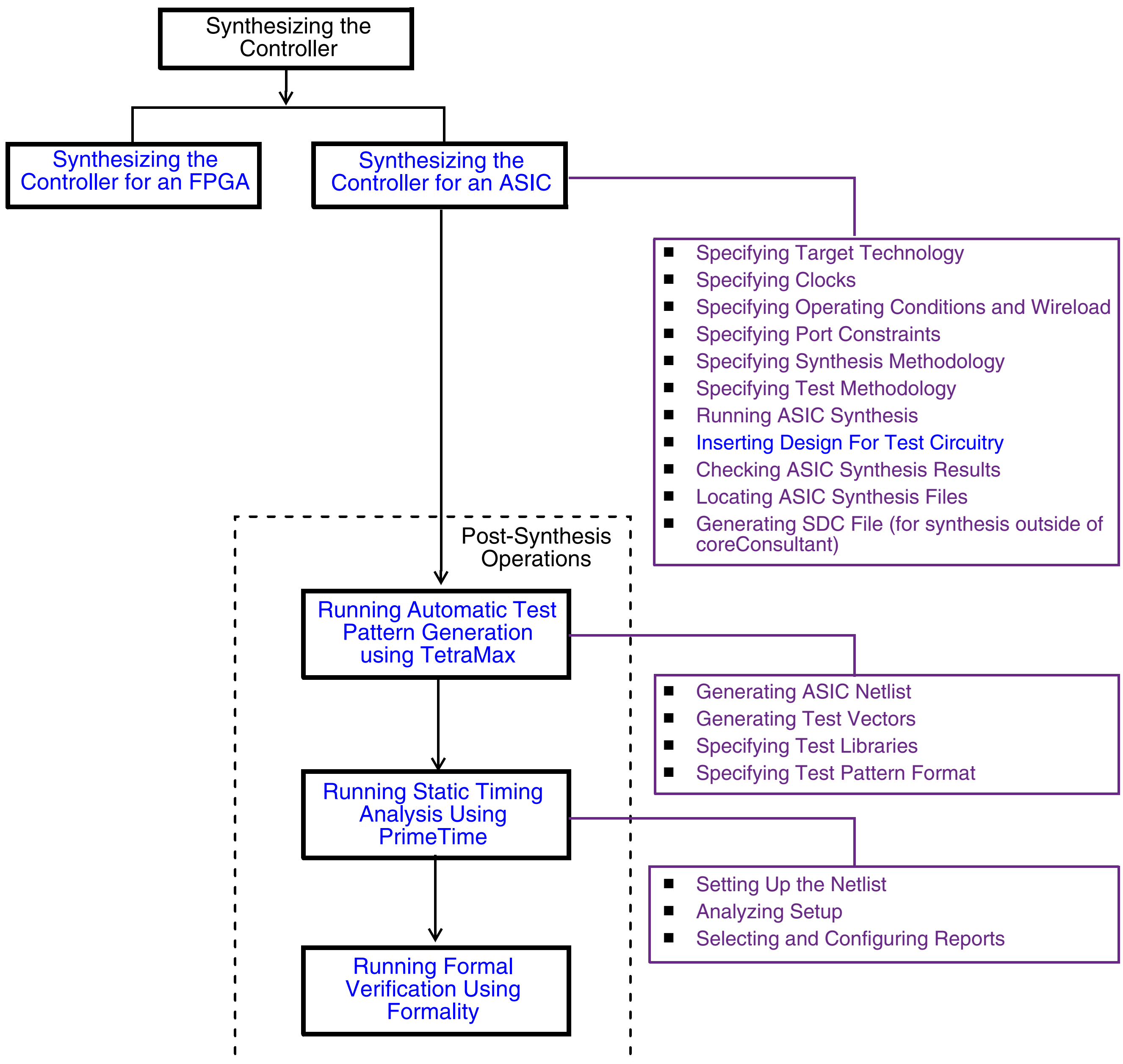

Table 1-6 provides controller ASIC and FPGA synthesis tasks and the respective documentation locations:

Table 1-6 Controller Synthesis Tasks

Task	Description	Document Location
Synthesizing the Controller for an ASIC	Map and synthesize the controller to an ASIC using the Synopsys Design Complier tool within the coreConsultant tool and check the results.	■ "Synthesizing the Controller for an ASIC" in Chapter 4 of *User Guide* ■ "Synthesis Flow" in Appendix B of *Databook*
Synthesizing the Controller for an FPGA	Perform system-level FPGA synthesis and implementation. **Note:** Unit-level FPGA synthesis using coreConsultant is not recommended for the DWC_usb3 controller.	*User Guide:* ■ "Synthesizing the Controller for an FPGA" in Chapter 4 ■ Appendix B, "FPGA Implementation of DWC_usb3 Controller"
Inserting Design For Test Circuitry	Perform DFT insertion during synthesis.	"Inserting Design For Test Using Design Compiler" in Chapter 4 of *User Guide*
Post-Synthesis Operations		
Running Automatic Test Pattern Generation using TetraMax	Perform Automatic Test Pattern Generation (ATPG) using the Synopsys Tetramax tool. **Note:** This activity is not applicable to FPGA implementations.	"Running Automatic Test Pattern Generation using TetraMax" in Chapter 4 of *User Guide*
Running Static Timing Analysis Using PrimeTime	Perform Static Timing Analysis (STA) by using the Synopsys PrimeTime tool. **Note:** This activity is not applicable to FPGA implementations.	"Running Static Timing Analysis" in Chapter 4 of *User Guide*
Running Formal Verification Using Formality	Use the Formal Verification activity for RTL-to-gate comparisons in coreConsultant. **Note:** This activity is not applicable to FPGA implementations.	"Performing Formal Verification" in Chapter 4 of *User Guide*

1.6 Integrating the Controller with the PHY

Figure 1-4 shows the usage flow for integration tasks and subtasks of the DWC_usb3 controller.

Figure 1-4 Controller Integration Tasks

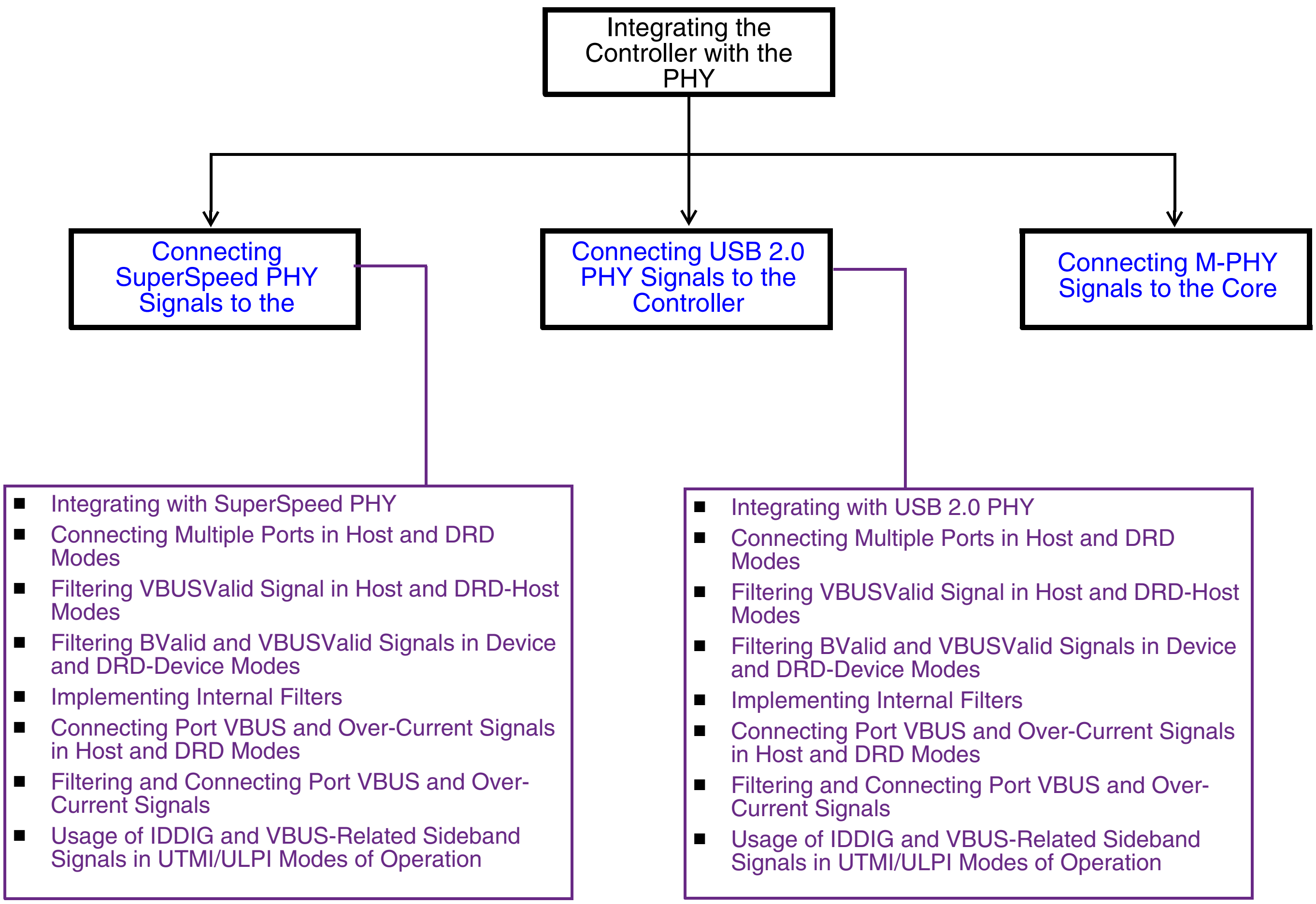

Table 1-7 provides controller-PHY integration tasks and the respective documentation locations:

Table 1-7 Controller-PHY Integration Tasks

Task	Description	Document Location
Connecting SuperSpeed PHY Signals to the Controller	Integrate the controller with SuperSpeed PHY	Chapter 5 of User Guide ■ "Integrating with SuperSpeed PHY" ■ "Connecting Multiple Ports in Host and DRD Modes" ■ "Filtering VBUSValid Signal in Host and DRD-Host Modes" ■ "Filtering BValid and VBUSValid Signals in Device and DRD-Device Modes" ■ "Implementing Internal Filters" ■ "Connecting Port VBUS and Over-Current Signals in Host and DRD Modes" ■ "Filtering and Connecting Port VBUS and Over-Current Signals" ■ "Usage of IDDIG and VBUS-Related Sideband Signals in UTMI and ULPI Modes" Reference: ■ "USB 3.0 PIPE3 PHY Interface" in Chapter 7of *Databook* ■ Application Notes - DesignWare Cores SuperSpeed USB 3.0 Controller and PHY Simulation Debug Guide - DesignWare Cores USB3.0 PHY – Controller Integration Guide

Table 1-7 Controller-PHY Integration Tasks (Continued)

Task	Description	Document Location
Connecting USB 2.0 PHY Signals to the Controller	Integrate the controller with USB 2.0 PHY if you select “High-Speed PHY Interface(s)?” configuration parameter to a non-zero value.	Chapter 5 of User Guide ■ Integrating with USB 2.0 PHY ■ “Connecting Multiple Ports in Host and DRD Modes” ■ “Filtering VBUSValid Signal in Host and DRD-Host Modes” ■ “Filtering BValid and VBUSValid Signals in Device and DRD-Device Modes” ■ “Implementing Internal Filters” ■ “Connecting Port VBUS and Over-Current Signals in Host and DRD Modes” ■ “Filtering and Connecting Port VBUS and Over-Current Signals” ■ “Usage of IDDIG and VBUS-Related Sideband Signals in UTMI and ULPI Modes” Reference: ■ USB 2.0 UTMI+ (Level 3) PHY Interfaces in Chapter 7 of *Databook* - “USB 2.0 UTMI+ Parallel Interface Signals” - “USB 2.0 UTMI+ Vendor Control Interface Signals” - “USB 2.0 UTMI+ OTG Interface Signals” - “LPM Interface Signals” ■ USB 2.0 ULPI PHY Interface - “USB 2.0 ULPI PHY Interface Signals” in Chapter 7 of *Databook* ■ Application Notes - DesignWare Cores SuperSpeed USB 3.0 Controller and PHY Simulation Debug Guide - DesignWare Cores USB3.0 PHY – Controller Integration Guide

1.7 Integrating the Controller with Application RTL (SoC)

Figure 1-5 shows the usage flow for integrating the DWC_usb3 controller with application RTL.

Figure 1-5 Controller-SoC Integration Tasks

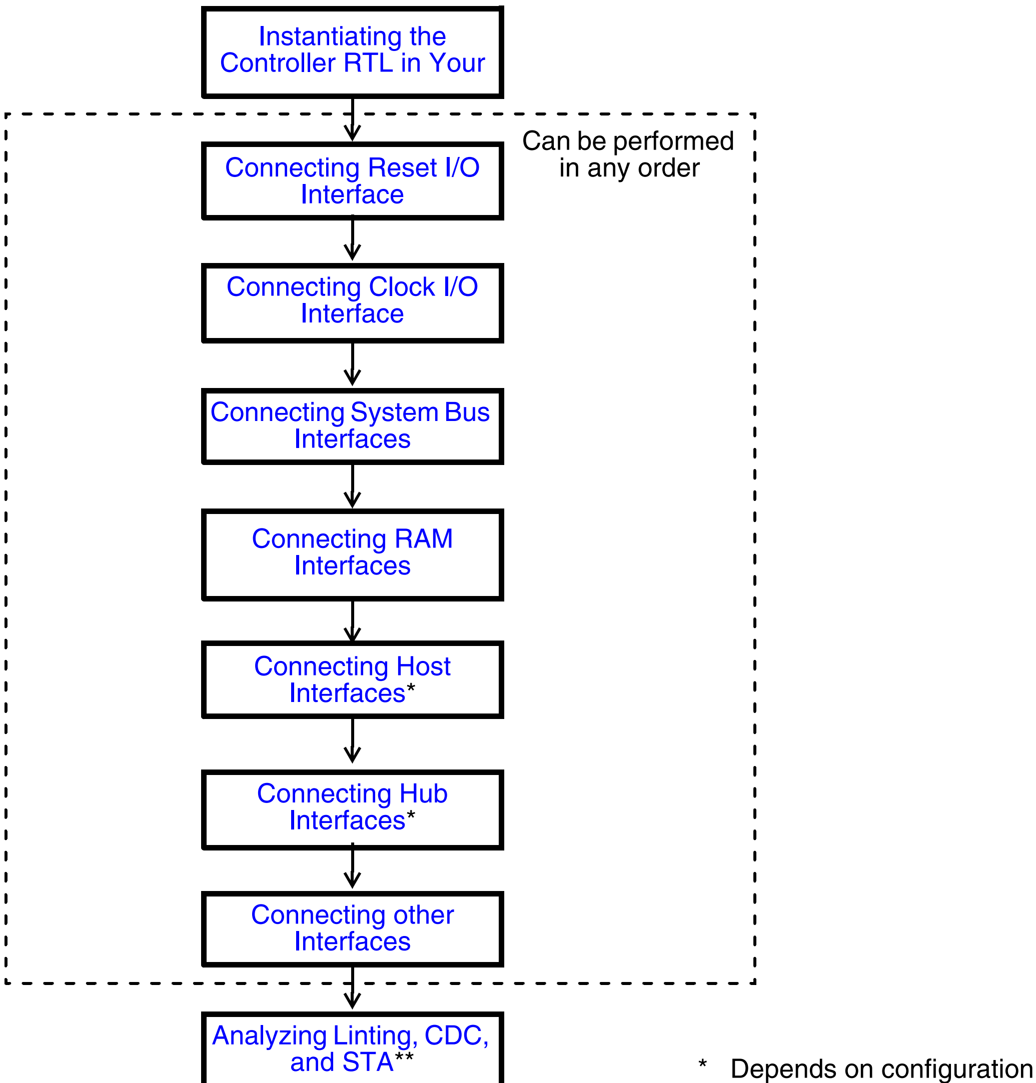

Table 1-8 includes the controller-SoC integration tasks and the respective documentation locations:

Table 1-8 Controller-SoC Integration Tasks

Task	Description	Documentation Location
Instantiating the Controller RTL in Your Design	Instantiate the top-level module of the controller that is generated in design.	■ Chapter 5 of *User Guide* - "Generating an Example Controller Instantiation" - "Integration Examples"
Connecting Reset I/O Interface	Check the reset requirements for DWC_usb3.	■ *Databook* - "Reset Generation" in Chapter 2 - "Asynchronous Resets" in Chapter 2 - "Software Resets" in Chapter 2 - "System Clock, Reset, and Control Signals" in Chapter 7
Connecting Clock I/O Interface	Check the clock requirements for DWC_usb3,	■ *Databook* - "Clock Generation and Clock Tree Synthesis (CTS) Requirements" in Chapter 3 - "System Clock, Reset, and Control Signals" in Chapter 7
Connecting System Bus Interfaces	Connect the controller's system bus interface to the system interfaces: ■ AHB Master Interface ■ AHB Slave Interface ■ Native Master Interface ■ Native Slave Interface ■ AXI Master Interface ■ AXI Slave Interface	■ System Bus Interface Overview in *Databook* ■ Chapter 7 of *Databook*: - "AHB Master Interface Signals" - "AHB Slave Interface Signals" - "Native Master Interface Signals" - "Native Slave Interface Signals" - "AXI Master Interface Signals" - "AXI Slave Interface Signals"
Connecting RAM Interfaces	Connect the RAM interface to the system interfaces	"RAM Interface Signals" in Chapter 7 of *Databook*
Connecting Host Interfaces	Connect the host interface to the system interfaces	"Host Interface Signals" in Chapter 7 of *Databook*
Connecting Hub Interfaces	Connect the hub application interface to the system interfaces	"Hub Interface Signals" in Chapter 7 of *Databook*
Connecting other Interfaces	Connect the application interface to the system interfaces: ■ ADP Interface ■ ACA Interface ■ Miscellaneous Signal Interface ■ Synopsys Test Environment Interface"	Chapter 7 of *Databook*: ■ "ADP Interface Signals" ■ "ACA Interface Signals" ■ "Miscellaneous Signal Interface Signals" ■ "Synopsys Test Environment Interface Signals" ■ "JTAG Interface Signals"

Table 1-8 Controller-SoC Integration Tasks (Continued)

Task	Description	Documentation Location
Analyzing Linting, CDC, and STA	Analyze Linting, CDC, and STA that are part of the coreConsultant flow.	■ For Lint and CDC, - Appendix B, "Clock Domain Crossing" in *Databook* - "Running Spyglass Lint and CDC" in Chapter 2 of *User Guide* ■ For STA, - "Running Static Timing Analysis using PrimeTime" in Chapter 4 of *User Guide* - "Synthesis Flow" in Appendix B of *Databook*

1.8 Programming the DWC_usb3 Controller

Table 1-9 provides details on where to find information for programming the DWC_usb3 controller.

Table 1-9 Programming Tasks

Task	Related Documentation
Programming the device	■ "Programming DWC_usb3 in Device Mode" in *Programming Guide* ■ "Device Data Structure" in *Programming Guide*
Programming the host	■ "Programming DWC_usb3 in Host Mode" in *Programming Guide* ■ xHCI Specification
Programming the controller when Hibernation is enabled	"Programming DWC_usb3 for Hibernation" in *Programming Guide*

1.9 Using/Implementing DWC_usb3 Features in Your Design

Table 1-10 lists features of the DWC_usb3 controller and the areas in the documentation where they are discussed in more detail.

Table 1-10 Related Documentation for DWC_usb3 Features

Feature/Task	Description	Related Documentation
Hibernation	Hibernation is a process in which the DWC_usb3 controller, on software request or on detecting suspend or P3 from the PHY, saves its internal state and enters low power mode. In low power mode, software can turn off the power supply to the controller. On detecting wakeup conditions, the controller restores its the internal states and enters normal operational state.	■ "Hibernation" in *Databook* ■ "Programming DWC_usb3 for Hibernation" in *Programming Guide*
Battery Charger	Battery charging is the ability to charge devices using USB.	■ "Battery Charger" in *Databook*
External Buffer Control	Device hardware External Buffer Control (EBC) is a minimized hardware-level control interface that allows the application to throttle the DMA read/write of data without involving software intervention.	■ "External Buffer Control" in *Databook*
xHCI Debug Capability	The xHCI Debug Capability (DbC) allows one of the host ports to negotiate as an upstream (device) port.	■ "xHCI Debug Capability" in *Programming Guide*
High-Speed InterChip (HSIC)	A chip-to-chip HS interconnect within a product.	■ "High Speed Inter-Chip (HSIC) Feature" in *Databook* ■ "HSIC Interface Signals" in Chapter 7 of *Databook*

1.10 Customer Support

For customer support:

- Enter a call through SolvNet.
  - Go to http://solvnet.synopsys.com/support/open_case.action.
    and provide the requested information, including:
    - Product: DesignWare Cores
    - Sub Product: USB 3.0 Device, USB 3.0 Host, USB 3.0 Hub, or USB 3.0 DRD
    - Version: 3.30b
    - Subject: USB 3.0
    - Problem Type:
    - Priority:
    - Title: <insert DWC_usb3>
    - Description: For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
- Send an e-mail message to support_center@synopsys.com.
  - Include the Product name, Sub Product name, and Version (product release number) in your e-mail so it can be routed correctly.
  - Provide the following additional information, if applicable:
    - For environment setup problems, run the debug_info command in the coreConsultant GUI and include the text file generated.
    - For configuration failures, include the error messages that appear in the coreConsultant GUI console pane.
    - For simulation failures, include a text file with your specific configuration. Generate this text file using the coreConsultant GUI's write_batch_script command. Also include the log file from the <workspace>/simulation/ directory, the log file for the specific test, and a VPD/VCD waveform dump file.
    - For synthesis failures, include the log file from the <workspace>/syn/directory.
- Telephone your local support center.
  - North America:

    Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
  - All other countries:

http://www.synopsys.com/Support/GlobalSupportCenters